



Basics of
Probability

Finite
probability space

Discrete
distribution

Geometric way
to calculate
probability

Conditional
probability

Independence of
events

Lecture notes for Applies statistics, Probability theory

Lecture 2.

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Presentation Overview

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Example of probability

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Imagine you repeat the same experiment numerous times. You do not necessarily expect to see the same result each time. Some results might occur more frequently than others. We account for this tendency using probability. To do so, we need to be clear about what results an experiment can have. For example, you flip a coin. We might agree that the only possible results are a head or a tail, thus ignoring the possibilities that (say) a bird swoops down and steals the coin; the coin lands and stays on edge; the coin falls between the cracks in the floor and disappears; and so on. By doing so, we have idealized the experiment.



Definition

Consider a fixed experiment K . Those outcomes of the experiment, which we can not divide into smaller parts are called elementary events. The elementary events are generally denoted by ω (Greek omega).

The set of all elementary events is called the sample space (probability space).

Notation: Ω (Greek Omega)

Examples

- ① Toss a coin. $\Omega = \{H, T\}$
- ② Toss two coins. $\Omega = \{HH, HT, TH, TT\}$
- ③ Roll a die. $\Omega = 1, 2, 3, 4, 5, 6$
- ④ Choose a point from the interval $(0, 1)$. Then $\Omega = (0, 1)$



The subsets of Ω are called events.

Example

- ① Toss two coins. Let A denote that we obtain at least one H . Then $A = \{HH, HT, TH\}$.
 - ② Roll a die. Let A denote that the result is even. Then $A = \{2, 4, 6\}$.
- Ω is called the sure event. $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$
 - $\emptyset$ is called the impossible event. $\emptyset = \{ \}$



Operation on events

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- $A \text{ or } B = A \cup B = A + B$
- $A \text{ and } B = A \cap B = A \cdot B$
- opposite of $A = \text{not } A = \bar{A}$
- the difference of A and $B = A \setminus B$
- precisely one of A and B occurs = symmetric difference of A and $B = A \circ B$
- A implies B means that $A \subseteq B$



Relative frequency

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Definition

Repeat the experiment n times. The event A occurs k_A times. Then

$$\frac{k_A}{n}$$

is called the relative frequency of A .

Theorem

The relative frequency is stable if n is large:

$$\frac{k_A}{n} \sim P(A),$$

where $P(A)$ is a fixed number.



Axioms

- ① Non-negative: $P(A) \geq 0$, for any event A .
- ② Normed: $P(\Omega) = 1$.
- ③ Countably additive:

$$P(A_1 + A_2 + \dots) = P(A_1) + P(A_2) + \dots,$$

if $A_1, A_2, \dots$ are pairwise exclusive.



Finite probability space

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Definition

$p_1, \dots, p_N$ is called a finite (probability) distribution, (or mass function), if the numbers p_i are non-negative and

$$\sum_{i=1}^N p_i = 1.$$

Definition

Any N element probability space can be described as follows.

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_N\},$$

$$P(A) = \sum_{\omega_i \in A} p_i,$$

for any event A , where $p_1, \dots, p_N$ is a given distribution.



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Definition

Assume that the outcomes of the experiment are equally possible. Then $p_1 = p_2 = \dots = p_N$. As $\sum_{i=1}^N p_i = 1$, therefore

$$p_1 = p_2 = \dots = p_N = \frac{1}{N},$$

and

$$P(A) = \sum_{\omega_i \in A} p_i = \sum_{\omega_i \in A} \frac{1}{N}.$$

So we obtain the classical rule to calculate probability

$$P(A) = \frac{\text{number of elements of } A}{\text{number of elements of } \Omega} = \frac{\text{number of favorable outcomes}}{\text{number of all outcomes}}.$$

This is the classical rule, which can be applied to solve several but not all problems.



Concept Question

Ten people, 5 women and 5 men are sitting around a round table. Find the probability, that neither two women nor two men are sitting next to each other.



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Concept Question

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$$P(A) = \frac{4! \cdot 5!}{9!}$$



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Concept Question

From a deck of cards three cards are dealt. Find the probability that there isn't any spade among them.



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Concept Question

From a deck of cards three cards are dealt. Find the probability that there isn't any spade among them.

$$P(A) = \frac{C_{39}^3}{C_{52}^3}$$



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Concept Question

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Concept Question

Find the probability that on the lottery 5 from 90 we hit at least three winning numbers.



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Find the probability that on the lottery 5 from 90 we hit at least three winning numbers.

$$P(A) = \frac{C_5^3 C_{85}^2 + C_5^4 C_{85}^1 + C_5^5 C_{85}^0}{C_{90}^5}$$



Discrete distribution

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$p_1, p_2, \dots$ is called a discrete (probability) distribution, if the numbers p_i are non-negative and

$$\sum_{i=1}^{\infty} p_i = 1.$$



Geometric way to calculate probability

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Let G be a subset of $\mathbb{R}^n$ ($n = 1, 2, 3$). Choose a point uniformly at random from G . Let $A \subseteq G$. Then the probability that the point is inside A is

$$P(A) = \lambda(A)/\lambda(G),$$

where λ is the length, area or volume if we are on the line, plane or space, respectively (we assume $0 < \lambda(G) < \infty$).



Concept question: Monty Hall problem

Basics of Probability

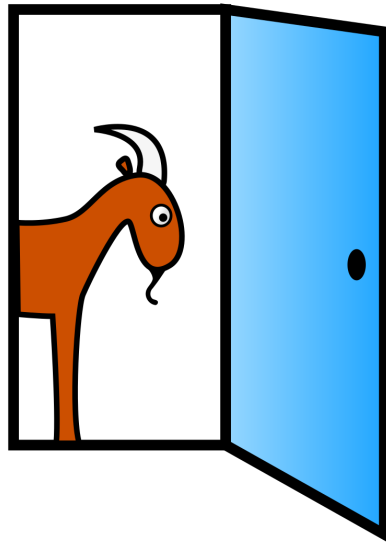
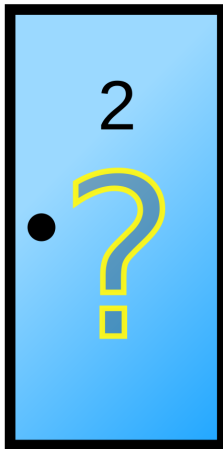
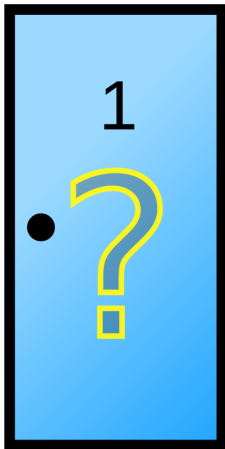
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Conditional probability

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Definition

Given two events A and B , with $P(B) > 0$, the conditional probability of A given B ($P(A|B)$) is the probability of A occurring if B has happened. A is assumed to be the set of all possible outcomes of an experiment. The conditional probability can be found by the quotient of the probability of the intersection of events A and B , that is, $P(A \cap B)$, the probability at which A and B occur together, and the probability of B :

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



Total probability theorem

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Definition

The sequence of events $A_1, A_2, \dots$ is called a partition, if they are pairwise exclusive and their union is the whole sample space. Obviously

$$P(A_1) + P(A_2) + \dots = 1.$$

Theorem (Total probability)

Let $B_1, B_2, \dots$ be a partition of the sample space. Assume that $P(B_i) > 0$ for all i . Then for any event A we have

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots.$$



Bayes' formula

Let A and B have positive probabilities. Then $P(B|A) = P(A|B)P(B)/P(A)$.

Bayes' theorem

Let A be an event, $B_1, B_2, \dots$ a partition of the sample space, $P(A) > 0, P(B_i) > 0, i = 1, 2, \dots$. Then for any i we have

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^{\infty} P(A|B_j)P(B_j)}.$$



Independence of two events

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Definition

We say that A is independent of B if the occurrence of B does not affect the probability of A . That is

$$P(A|B) = P(A).$$

However, this definition is not symmetric and we should assume that $P(B) > 0$. In the following definition the roles of A and B are symmetric.

Definition

We say that A and B are independent if

$$P(AB) = P(A)P(B).$$



Independence of multiple events

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Definition

We say that the events $A_1, A_2, \dots$ are pairwise independent if any two of them are independent, that is

$$P(A_i A_j) = P(A_i)P(A_j), i \neq j.$$

Definition

We say that the events $A_1, A_2, \dots, A_n$ are independent if for any $k = 1, 2, \dots, n$ and for any combinations $i_1, \dots, i_k$ of the numbers $1, 2, \dots, n$ we have

$$P(A_{i_1} A_{i_2} \dots A_{i_k}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_k}).$$